

BDM Belief Elicitation and Behavioral Incentive Compatibility

Design review

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Danz, Vesterlund, and Wilson (JEP 2024 and forthcoming WP) mentions BDM fails BIC in pure incentive test: 69% of subjects pick $q = 0$ under p-BDM at induced $\theta = 0.2$. No public WP available.

Chakraborty and Kendall (2025) formalize UJS. BDM admits many justifiable non-truthful reports; MPL admits one, may be better BIC.

Research questions and contribution

Research questions:

1. Why does BDM belief elicitation fail behavioral IC?
2. Does MPL perform better, despite theoretical equivalence?
3. Does the presentation format interact with task complexity for belief elicitation?

Contribution: identify the mechanism behind BDM's failure, test UJS's design implication empirically in belief elicitation, and measure the complexity-by-mechanism interaction.

Theoretical IC: 2 Approaches

- ▶ Karni (2009): Dominance + Probabilistic Sophistication
- ▶ Azrieli et al. (2018): Monotonicity

Azrieli's monotonicity is strictly weaker — the minimal sufficient assumption for IC.

Hypotheses

- H1. BDM fails behavioral IC, via two weak conditions (DVW 2022):
 - H1a. Explaining BDM's incentives does not increase accuracy.
 - H1b. Subjects do not choose the payoff-maximizer under pure incentives.
- H2. Belief MPL achieves better behavioral IC than single-report BDM, despite theoretical equivalence, because MPL admits a unique justifiable action per row (UJS) while BDM admits many.
- H3. The BDM-MPL truthful reporting gap increases with task complexity.

Experimental design

Ball and urn paradigm.

T#	Mechanism	Instructions	Identifies
1	Single-report BDM	Full info + quiz	H1 baseline
2	Belief MPL	Full info + quiz	H2, format matters
3	Single-report BDM	No info	H1a: info/no-info
4	Pure-incentives test	—	H1b: direct test
5	Flat fee	Accuracy encouragement	No-mechanism benchmark

Within-subject: priors (easy) and posteriors after signals (hard). Identifies H3.

Outcomes: Accuracy at $\varepsilon = 0$ and $\varepsilon = 5\text{pp}$ (DVW 2022). Also multi-switching, response time, comprehension.

Question 1: the overall design

Is there anything I am missing?

- ▶ Is the current design sufficient to identify H1 through H3?
- ▶ How to design p-BDM pure incentive test? (coming up next)
- ▶ Other considerations?

p-BDM pure-incentives test (H1b)

DVW report one p-BDM pure incentive test result: 69% of subjects pick $q = 0$ at $\theta = 0.2$. WP is not available.

- ▶ $q = 0$ pays with probability 0.5 regardless of event.
- ▶ $q = \theta = 0.2$ (the maximizer) pays with probability 0.68 if E , 0.48 otherwise, for an expected 0.52.
- ▶ The EV gap between the maximizer and the event-independent option is 2 percentage points.

Proposal A: Example at $\theta = 0.2$

$$P(\text{win} \mid E, q) = q + (1 - q^2)/2, \quad P(\text{win} \mid \neg E, q) = (1 - q^2)/2.$$

Report q	$P(\text{win} \mid E)$	$P(\text{win} \mid \neg E)$	EV	
0.0	0.500	0.500	0.500	event-independent
0.1	0.595	0.495	0.515	
0.2	0.680	0.480	0.520	maximizer, $q^* = \theta$
0.3	0.755	0.455	0.515	
0.4	0.820	0.420	0.500	
0.5	0.875	0.375	0.475	
0.6	0.920	0.320	0.440	
0.7	0.955	0.255	0.395	
0.8	0.980	0.180	0.340	
0.9	0.995	0.095	0.275	
1.0	1.000	0.000	0.200	

Mechanism and q values hidden. Subject compares event-contingent payoff pairs directly — no contingent reasoning across r is required.

Proposal B: Example at $\theta = 0.2$

Urn: 20 red balls, 80 blue. Event $E = \text{red drawn}$. $P(E) = 0.2 = \theta$.

Subject sees:

You will choose a number q from $\{0.0, 0.1, 0.2, \dots, 1.0\}$. A random r is drawn uniformly from $[0, 1]$.

- ▶ *If $q \geq r$: you win $\$H$ if red.*
- ▶ *If $q < r$: you win $\$H$ with probability r .*

Which q do you pick?

No payoff pairs shown. To find the maximizer, the subject must reason through what happens at each r given each candidate q — full contingent reasoning demand preserved.

Question 2: A, B, or something else?

The pure-incentives test diagnoses BIC failure by asking whether subjects identify the payoff-maximizer with beliefs stripped from the task.

A and B differ in how much of the p-BDM structure remains:

- ▶ A: mechanism hidden; subjects see only event-contingent payoff pairs.
- ▶ B: native p-BDM framing; contingent reasoning across r preserved.

Both designs are separate between-subject arms, with θ varying within-subject. Running A and B within-subject isolates menu-level vs. contingent-reasoning failure.

A, B, or something else?

MPL format: the problem

Brown and Healy (2018): MPL is theoretically IC, but presentation format determines whether it holds in practice.

How list format can break IC:

1. All rows visible on one screen $\Rightarrow$ compound RPS structure is salient.
2. Subjects treat the list as one decision; apply ROCL.
3. ROCL + non-EU preferences $\Rightarrow$ monotonicity violation (Segal 1990).
4. Monotonicity violated $\Rightarrow$ RPS no longer IC (Azrieli et al. 2018).

Fix: Separated format (one row per screen, random order): ROCL not triggered.

MPL format: options and tradeoffs

Format	Screens	ROCL concern	Burden	Precision
Full list (all rows on 1 screen)	1	Yes	High	1pp
Two-stage list (Holt & Smith)	2	Yes	Moderate	1pp
Full separated (100 rows)	100	No	High	1pp
Coarse separated	15–20	No	Moderate	5pp
Two-stage separated	20–30	No	High	1pp

Two-stage formats: the Multiple Switching dilemma

Two-stage MPL encounters a problem with how to handle multiple switching (MS):

- ▶ **Force single-switching in Stage 1.** Stage 2 implementable, but censors MS behavior.
- ▶ **Allow MS in Stage 1.** Stage 2 not implementable.
- ▶ **Skip Stage 2 for MS subjects.** Asymmetric rounds across subjects.

Applies to two-stage list and two-stage separated.

Question 3: MPL format

Proposal: coarse separated (15–20 rows, one per screen, random order) — the belief analog of UJS-E.

Pro: Works for our BIC test at 5pp precision

Con: does not generalize to precision-sensitive belief elicitation.

How to approach MPL format choice?